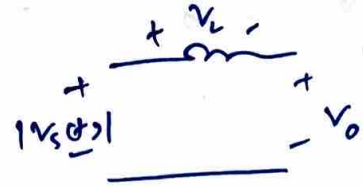
Week-2 Assignment
Numericals

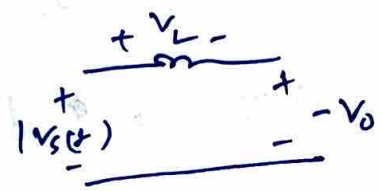
Q.2.

In $d(t)T_s$ duration, S_1 & S_4 is ON



$$\Rightarrow v_L = |v_s(t)| - v_o$$

In $(1-d(t))T_s$ duration, S_2 & S_3 is ON



$$\Rightarrow v_L = |v_s(t)| + v_o$$

Apply volt-sec

$$\frac{1}{T/2}\int_0^{T/2}\left(\left(|v_s(t)| - v_o\right)\cdot d(t)T_s + \left(|v_s(t)| + v_o\right)(1-d(t))T_s\right)\cdot dt = 0$$

Apply approximation

$$\Rightarrow \left(|v_s(t)| - v_o\right)d(t) + \left(|v_s(t)| + v_o\right)(1-d(t)) = 0$$

$$\Rightarrow |v_s(t)|d(t) - v_o d(t) + |v_s(t)| + v_o - |v_s(t)|d(t) - v_o d(t) = 0$$

$$2v_o d(t) = v_o + |v_s(t)|$$

$$\Rightarrow d(t) = \frac{\left(1 + \frac{|v_s(t)|}{v_o}\right)}{2} \longrightarrow \text{(D)}$$

Q.4.

The inductor ripple is maximum at

$$|v_s(t)| = \frac{v_o}{2} \qquad \text{(from Lec-10)}$$

$$\Rightarrow |v_s(t)| = \frac{400}{2} = 200 \longrightarrow (c)$$

Q.7.

apply volt-sec and approximations, (as in Q.2) we get

$$(|v_s(t)| + v_o)\, d(t) + (|v_s(t)| - v_o)(1 - d(t)) = 0$$

$$\Rightarrow \cancel{|v_s(t)|\, d(t)} + v_o\, d(t) + |v_s(t)| - v_o - \cancel{|v_s(t)|\, d(t)} + v_o\, d(t) = 0$$

$$\Rightarrow 2v_o\, d(t) = v_o - |v_s(t)|$$

$$\Rightarrow d(t) = \frac{v_o - |v_s(t)|}{2v_o}$$

at Positive peak $d(t) = 0.15$

$$\Rightarrow 0.15 = \frac{v_o - 325}{2v_o}$$

$$\Rightarrow 0.3v_o = v_o - 325$$

$$\Rightarrow \boxed{v_o = \frac{325}{0.7} = 464.28\,V}$$

Q.8.

$$L = \frac{V_o}{4 \cdot \Delta i_{L,max} \cdot f_{sw}}$$

lec - 10

$$I_{s,rms} \cdot V_{s,rms} = 1100$$

$$\Rightarrow I_{s,rms} = \frac{1100}{230}$$

$$= 4.78$$

$$\Rightarrow I_{s,pk} = 6.76A$$

$$\Rightarrow \Delta i_{L,max} = 10\% \text{ of } 6.76$$

$$= 0.676$$

$$\Rightarrow L = \frac{450}{4 \times 0.676 \times 60000}$$

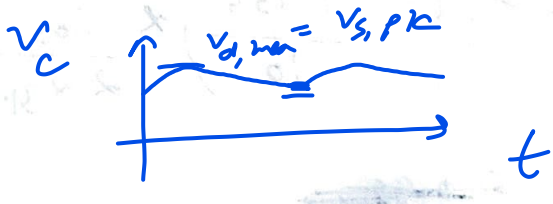
$$\boxed{L = 2.77 \text{ mH}}$$

Q.9.

from lec - 7.

$$V_{d,max} = V_{s,pk}$$

$$V_{d,min} = 0.9 \cdot V_{s,pk}$$



(as it is given ripple can be 10%).

$$\Rightarrow V_{d,min} = V_{d,max} \cos(\omega t_c)$$

$$\Rightarrow \omega t_c = \cos^{-1}\left(\frac{V_{d,min}}{V_{d,max}}\right)$$

$$\Rightarrow t_2 = \frac{\cos^{-1}(0.9)}{2\pi \times 50} = \frac{0.451}{2\times 3.14\times 50} \rightarrow \text{in radians}$$

$$\Rightarrow t_2 = 1.436\,ms$$

$$\Rightarrow C = \left(1 - \frac{t_2}{T/2}\right) \cdot \frac{P_L}{2 \cdot \Delta V_0 \cdot V_0} \cdot \frac{1}{f_s} \longrightarrow \text{lee (7)}$$

$P_L = 200\,W$

$$V_0 = \frac{V_{dy\,max} + V_{dy\,min}}{2} = \frac{220\times\sqrt{2} + 0.9\times 220\times\sqrt{2}}{2}$$

$$V_0 = 295.57$$

$$\Delta V_0 = V_{dy\,max} - V_{dy\,min}$$

$$= 220\times\sqrt{2} - 0.9\times 220\times\sqrt{2}$$

$$\Delta V_0 = 31.112$$

$$T = \frac{1}{50} = 20\,ms$$

$$\Rightarrow C = \left(1 - \frac{1.436}{10}\right) \times \frac{200}{2\times 31.112\times 295.57} \times \frac{1}{50}$$

~~$C = 0.86 \times$~~

$$C = 186.26\,\mu F$$

Q.10-

$$d(t) = 1 - \frac{|V_{s,pk} \sin \omega t|}{V_o}$$

$$\Rightarrow d(t)_{,min,180} = 1 - \frac{180 \times \sqrt{2}}{500} = 0.491$$

$$\Rightarrow d(t)_{,min,265} = 1 - \frac{265 \times \sqrt{2}}{500} = 0.250$$

Since, for varying input, the minimum $d(t)$ will be for 265V input, hence

$$\boxed{d(t)_{,min} = 0.25}$$